

Part A – Theory: Introduction to Regression in Machine Learning

1. Introduction to Regression

What is Regression in Machine Learning?

Regression is a supervised machine learning technique used to predict continuous numerical values based on input data. The objective of regression is to establish a relationship between independent variables (features) and a dependent variable (target). The model learns patterns from historical data and uses these patterns to estimate future or unknown values.

For example, regression models can predict house prices, stock market values, rainfall amounts, or students' examination scores.

Difference Between Regression and Classification

Although both regression and classification belong to supervised learning, they differ in the type of output they produce.

- **Regression** predicts continuous numerical values.
- **Classification** predicts discrete categories or classes.

For instance, regression may predict that a house will cost \$150,000, whereas classification may determine whether an email is “spam” or “not spam.”

Real-Life Examples

Regression Example: Predicting the selling price of a house based on factors such as location, size, and number of rooms.

Classification Example: Classifying whether a patient has diabetes (Yes/No) based on medical test results.

2. Types of Regression

2.1 Linear Regression

Basic Idea

Linear Regression is the simplest regression technique. It models the relationship between one independent variable and one dependent variable using a straight line.

The mathematical equation is:

$$[Y = \beta_0 + \beta_1 X + \epsilon]$$

Where:

- $\hat{Y}$ = predicted value
- X = input feature
- β_0 = intercept
- β_1 = slope
- ϵ = error term

The algorithm finds the best-fitting line that minimizes prediction errors.

Real-World Use Case

Predicting employee salary based on years of work experience.

Advantages

- Easy to understand and implement.
- Computationally efficient.
- Results are highly interpretable.

Limitations

- Assumes a linear relationship.
- Performs poorly on complex non-linear data.

- Sensitive to outliers.

2.2 Multiple Linear Regression

Basic Idea

Multiple Linear Regression extends simple linear regression by using multiple independent variables to predict one dependent variable.

The equation is:

$$[Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon]$$

The model estimates the contribution of several features simultaneously.

Real-World Use Case

Predicting house prices using factors such as area, number of bedrooms, age of the house, and location.

Advantages

- Uses multiple factors for prediction.
- Usually provides more accurate predictions.
- Helps analyze relationships among several variables.

Limitations

- More complex than simple linear regression.
- Can suffer from multicollinearity among predictors.
- Assumes linear relationships.

2.3 Polynomial Regression

Basic Idea

Polynomial Regression models non-linear relationships by adding polynomial terms (squared, cubed, etc.) of the independent variable.

The equation is:

$$[Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_n X^n + \epsilon]$$

Although the equation is polynomial, the model is still linear with respect to the coefficients.

Real-World Use Case

Modeling vehicle fuel consumption based on speed, where the relationship is often curved rather than linear.

Advantages

- Captures non-linear relationships.
- More flexible than linear regression.
- Can improve prediction accuracy when data patterns are curved.

Limitations

- High-degree polynomials may overfit data.
- Difficult to interpret.
- Computational complexity increases with degree.

3. Regression Metrics

Regression metrics evaluate how well a model predicts continuous values.

3.1 Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual and predicted values.

$$[MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|]$$

What it measures: Average prediction error without considering direction.

Lower MAE values indicate better model performance.

3.2 Mean Squared Error (MSE)

MSE computes the average of squared differences between actual and predicted values.

$$[\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2]$$

What it measures: Average squared prediction error.

Large errors receive higher penalties.

3.3 Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

$$[\text{RMSE} = \sqrt{\text{MSE}}]$$

What it measures: Average prediction error expressed in the original units of the target variable.

3.4 R² (Coefficient of Determination)

R² measures the proportion of variance in the dependent variable explained by the model.

$$[R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}]$$

Its value ranges from 0 to 1.

- (R² = 1): perfect prediction.
- (R² = 0): model explains no variance.

Comparison of Regression Metrics and Performance

Among the regression evaluation metrics, **R²** is considered the best metric for measuring how well a model explains the variability in the target variable. A higher R² value indicates better model performance because it means that the model can explain a larger proportion of the variance in the data. An R² value close to **1** represents excellent predictive performance, whereas an R² value close to **0** indicates poor performance.

For error-based metrics such as **MAE**, **MSE**, and **RMSE**, lower values indicate better model performance because they represent smaller differences between actual and predicted values.

Models with low MAE and RMSE are generally preferred since they produce more accurate predictions.

Among these metrics, **MAE** often provides the easiest interpretation because it expresses the average prediction error in the same units as the target variable. **RMSE** is widely used in practice because it penalizes large prediction errors more heavily than MAE, making it useful when large errors are undesirable.

In contrast, **MSE** is usually considered less interpretable because its values are expressed in squared units rather than the original units of the target variable. Although MSE is highly useful during model optimization, very large MSE values indicate poor predictive performance due to substantial prediction errors.

Overall, a regression model is considered to have **the best performance when it achieves a high R^2 value together with low MAE, MSE, and RMSE values**. Conversely, a model with **low R^2 and high error metrics demonstrates the lowest performance**, indicating that the model fails to accurately predict the target variable.

Comparison of Regression Metrics

Metric	Unit	Sensitivity to Large Errors	Meaning
MAE	Same as target variable	Low	Average absolute prediction error
MSE	Squared units	Very High	Average squared error
RMSE	Same as	High	Typical

Metric	Unit	Sensitivity to Large Errors	Meaning
	target variable		prediction error magnitude
R^2	Unitless	Not directly based on error magnitude	Proportion of variance explained

4. Underfitting and Overfitting

Underfitting

Underfitting occurs when a model is too simple to capture the underlying patterns in the data.

Consequently, it performs poorly on both training and testing datasets.

Example: Using a linear model for highly non-linear data.

Overfitting

Overfitting occurs when a model learns not only the true patterns but also noise and random fluctuations in the training data. As a result, it performs very well on training data but poorly on unseen data.

Causes of Overfitting in Polynomial Regression

- Using excessively high polynomial degrees.
- Small training datasets.
- Presence of noise in data.
- Too many model parameters.

Methods to Prevent Overfitting

1. **Cross-validation:** Evaluates model performance on different subsets of data.
2. **Regularization:** Techniques such as Ridge and Lasso penalize model complexity.
3. **Reduce model complexity:** Select a lower polynomial degree or fewer features.

5. Real-World Case Study

Healthcare: Predicting Diabetes Progression

A well-known healthcare study used multiple linear regression to predict disease progression in diabetic patients.

Goal

The objective was to predict the progression of diabetes one year after baseline medical measurements.

Data Used

Researchers collected clinical measurements from patients, including:

- Age
- Body Mass Index (BMI)
- Blood pressure
- Cholesterol levels
- Blood glucose measurements

Type of Regression Applied

Multiple Linear Regression was employed because several predictor variables influenced disease progression.

Key Results and Insights

The study found that factors such as BMI and blood glucose levels significantly affected diabetes progression. The regression model helped healthcare professionals identify high-risk patients and improve treatment planning.

This study demonstrated that regression techniques can provide valuable insights for medical decision-making and patient management.

References

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